CHAPTER 16 · DERIVATIONS

MAGNETIC FIELDS

5 derivations the STPM 2026 syllabus asks you to derive, deduce or show

1 FORCE ON A CURRENT-CARRYING CONDUCTOR

SYLLABUS 17.3(B)

RESULT $F = IlB \sin\theta$

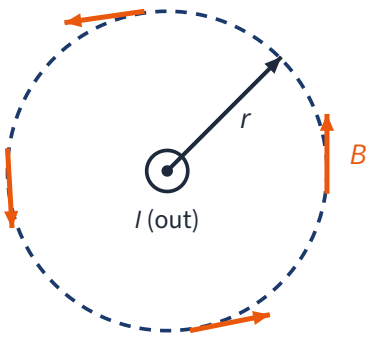
1. Each charge carrier q drifts with velocity v at angle θ to B: force $qvB \sin\theta$.
2. Number of carriers in length l: nAl.
3. Total force $F = (nAl)qvB \sin\theta$.
4. $I = nAqv$, so $\mathbf{F = IlB \sin\theta}$.

2 FIELD OF A LONG STRAIGHT WIRE (AMPÈRE'S LAW)

SYLLABUS 17.4(B)

RESULT $B = \dfrac{\mu_0 I}{2\pi r}$

1. Ampère's law: $\oint B \cdot dl = \mu_0 I$.
2. Choose a circle of radius r centred on the wire. By symmetry B is tangential and has the same size all round.
3. $\oint B \cdot dl = B(2\pi r)$.
4. $B(2\pi r) = \mu_0 I$, so $\mathbf{B = \dfrac{\mu_0 I}{2\pi r}}$.



3 FORCE BETWEEN TWO PARALLEL CONDUCTORS

SYLLABUS 17.5(A)

RESULT $F = \dfrac{\mu_0 I_1 I_2 l}{2\pi d}$

1. Field of wire 1 at wire 2 (distance d): $B_1 = \dfrac{\mu_0 I_1}{2\pi d}$, perpendicular to wire 2.
2. Force on length l of wire 2: $F = B_1 I_2 l$.
3. $\mathbf{F = \dfrac{\mu_0 I_1 I_2 l}{2\pi d}}$; an equal and opposite force acts on wire 1.
4. Currents in the same direction attract; opposite directions repel.

4 E/M IN THOMSON'S EXPERIMENT

SYLLABUS 17.6(C)

RESULT $\frac{e}{m} = \frac{E}{rB^2}$

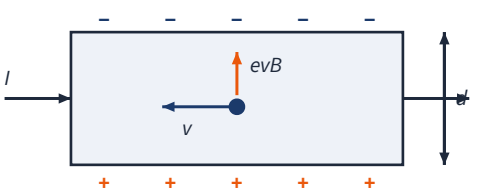
1. Crossed E and B fields are adjusted so the beam is undeflected: eE = evB, so $v = \frac{E}{B}$.
2. With E switched off, the electrons move in a circle of radius r in B: $evB = \frac{mv^2}{r}$.
3. $\frac{e}{m} = \frac{v}{rB}$.
4. Substitute $v = \frac{E}{B}$: $\mathbf{\frac{e}{m} = \frac{E}{rB^2}}$.

5 HALL VOLTAGE

SYLLABUS 17.7(B)

RESULT $V_H = \frac{BI}{net}$

1. Carriers (charge e, number density n) drift with velocity v through a slab of width d and thickness t, with B perpendicular.
2. The magnetic force evB pushes carriers to one face; charge builds up and sets up a Hall field $E_H = \frac{V_H}{d}$.
3. Equilibrium: $\frac{eV_H}{d} = evB$, so $V_H = Bvd$.
4. $I = neAv = ne(dt)v$, so $v = I/(nedt)$.
5. $\mathbf{V_H = \frac{BI}{net}}$.



B into page; V_H across *d*; thickness *t*